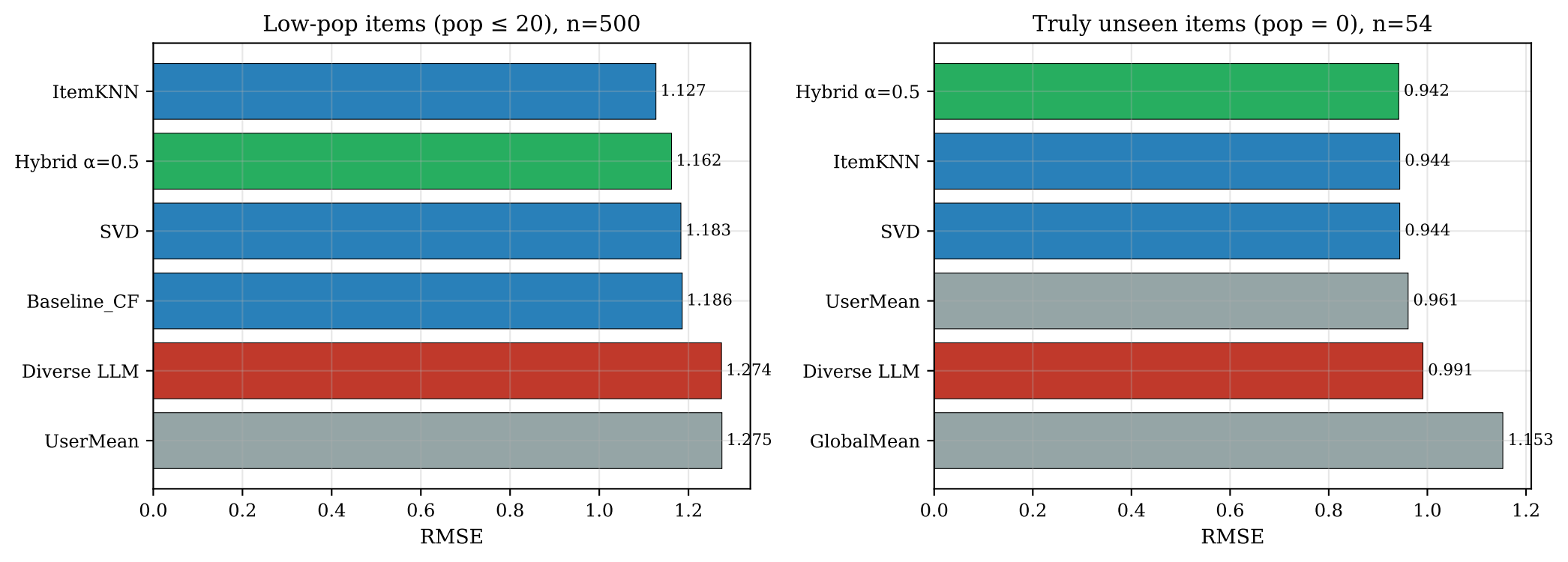
Low-pop items (pop ≤ 20), n=500
ItemKNN
Hybrid α=0.5
SVD
Baseline_CF
Diverse LLM
UserMean
1.127
1.162
1.183
1.186
1.274
1.275
0.0
0.2
0.4
0.6
0.8
1.0
1.2
RMSE
Truly unseen items (pop = 0), n=54
Hybrid α=0.5
ItemKNN
SVD
UserMean
Diverse LLM
GlobalMean
0.942
0.944
0.944
0.961
0.991
1.153
0.0
0.2
0.4
0.6
0.8
1.0
1.2
RMSE